LAPORAN PRAKTIKUM

POSTTEST (6)

ALGORITMA PEMROGRAMAN LANJUT

Disusun oleh:

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Kelas (A2 ‘25)

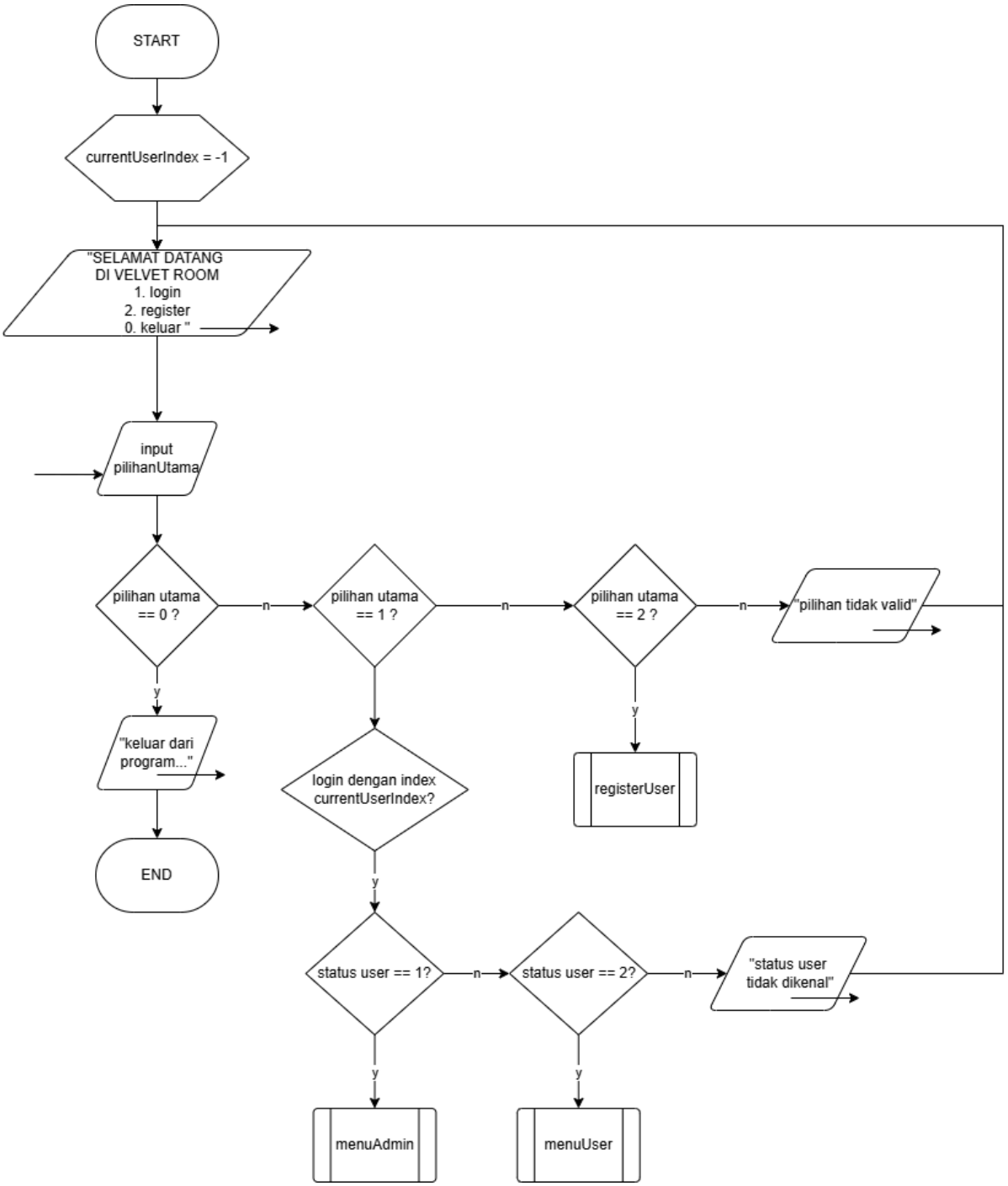
PROGRAM STUDI INFORMATIKA

UNIVERSITAS MULAWARMAN

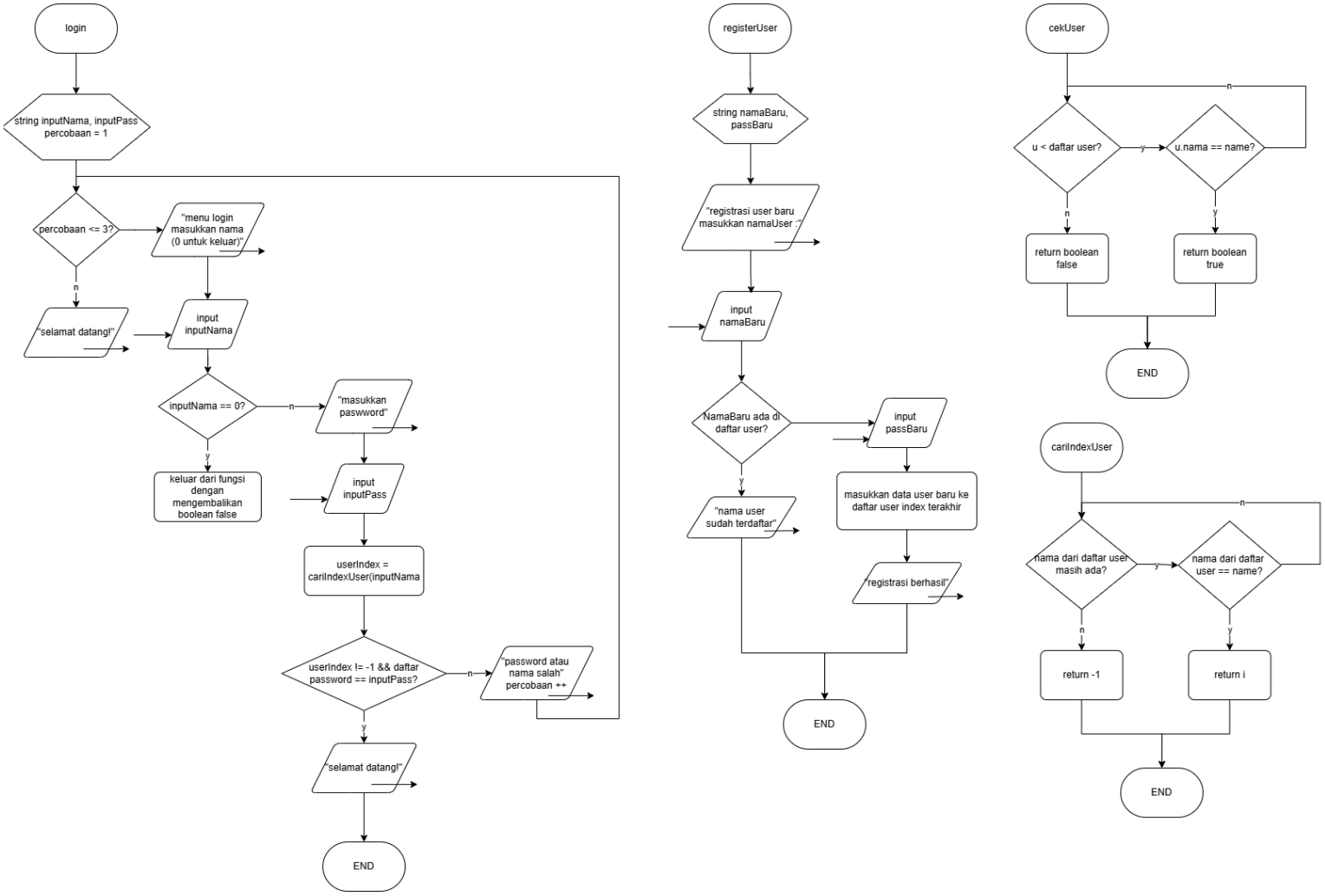
SAMARINDA

2025

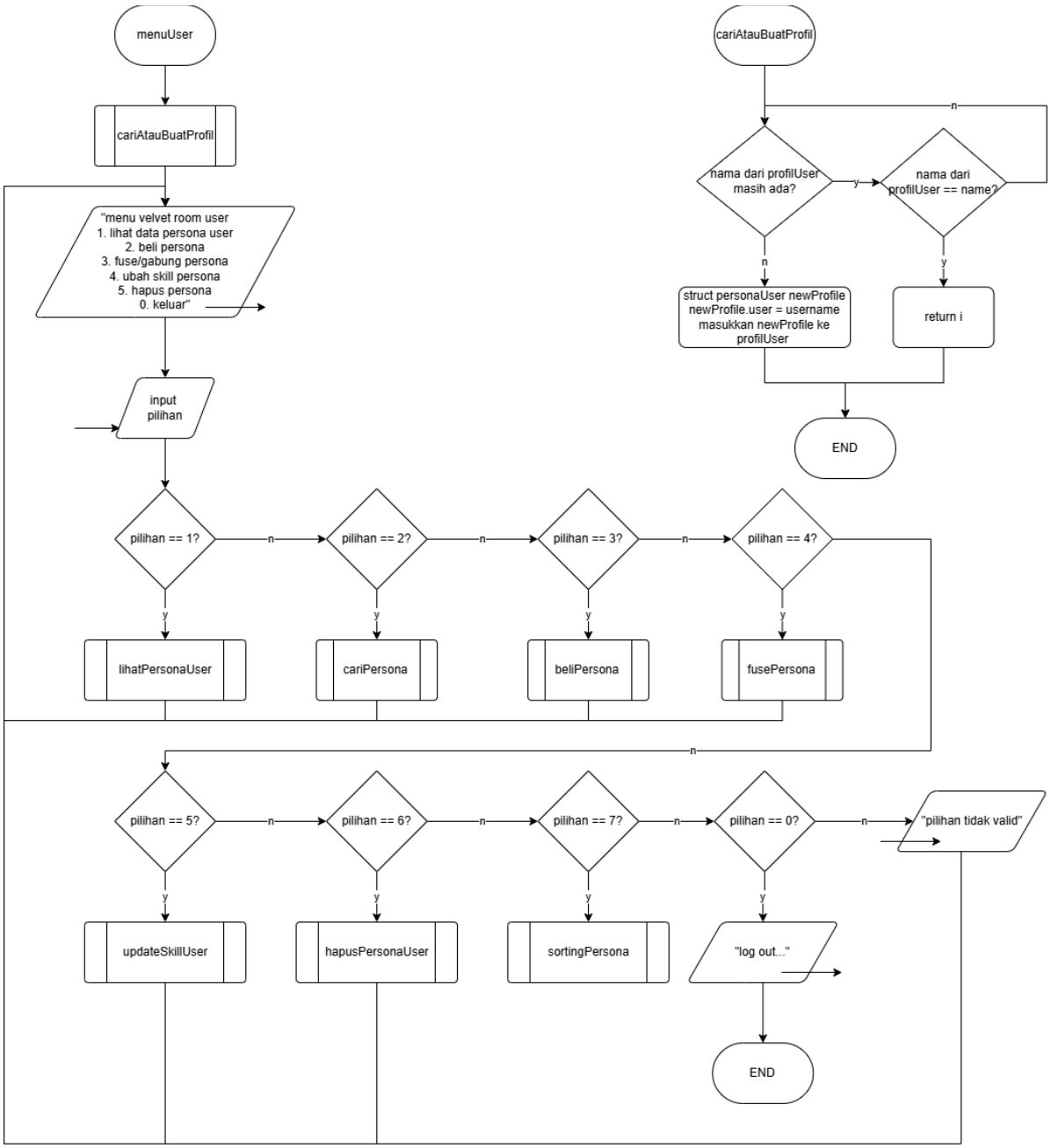
1. Flowchart



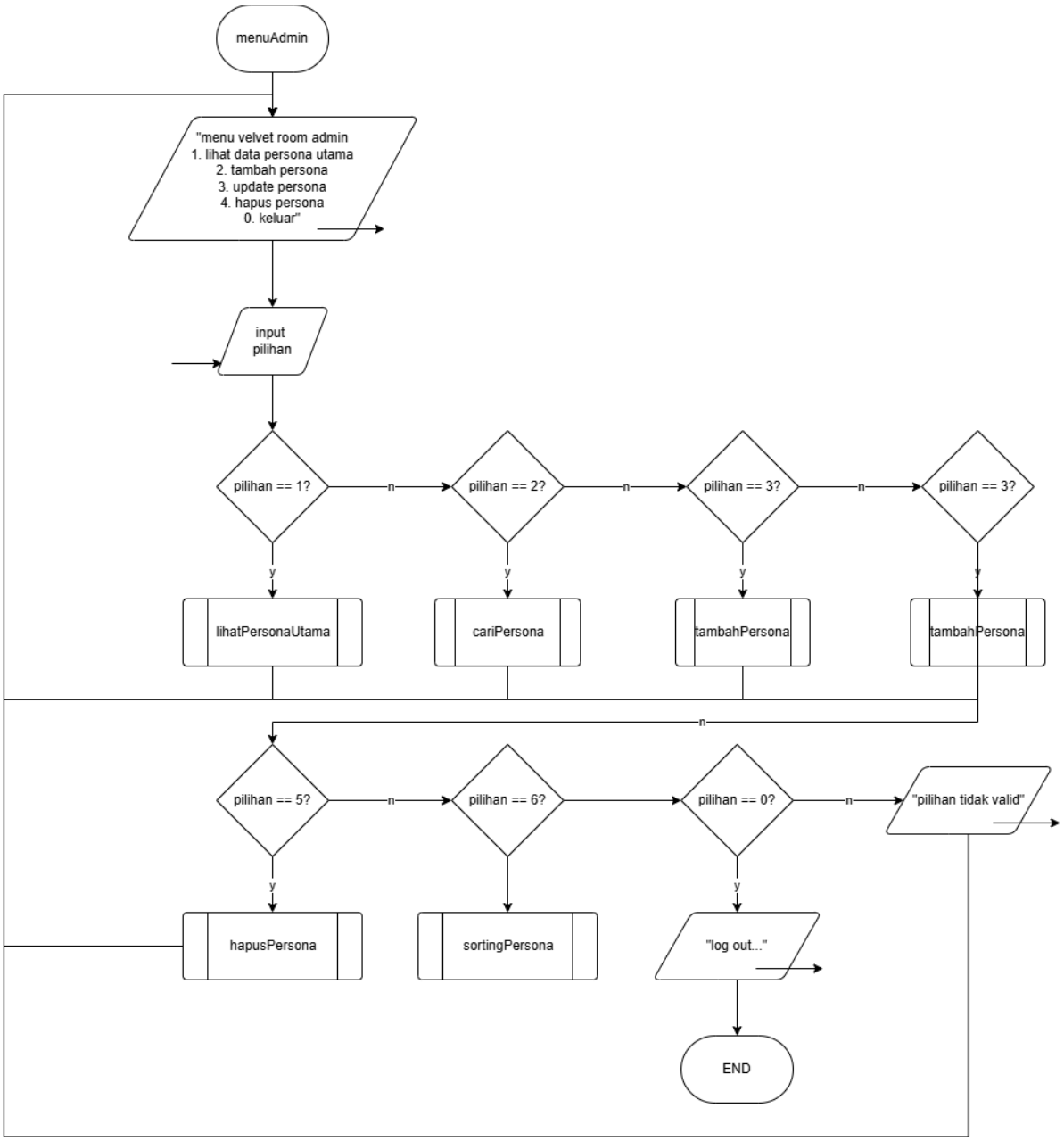
gambar 1.1 main program



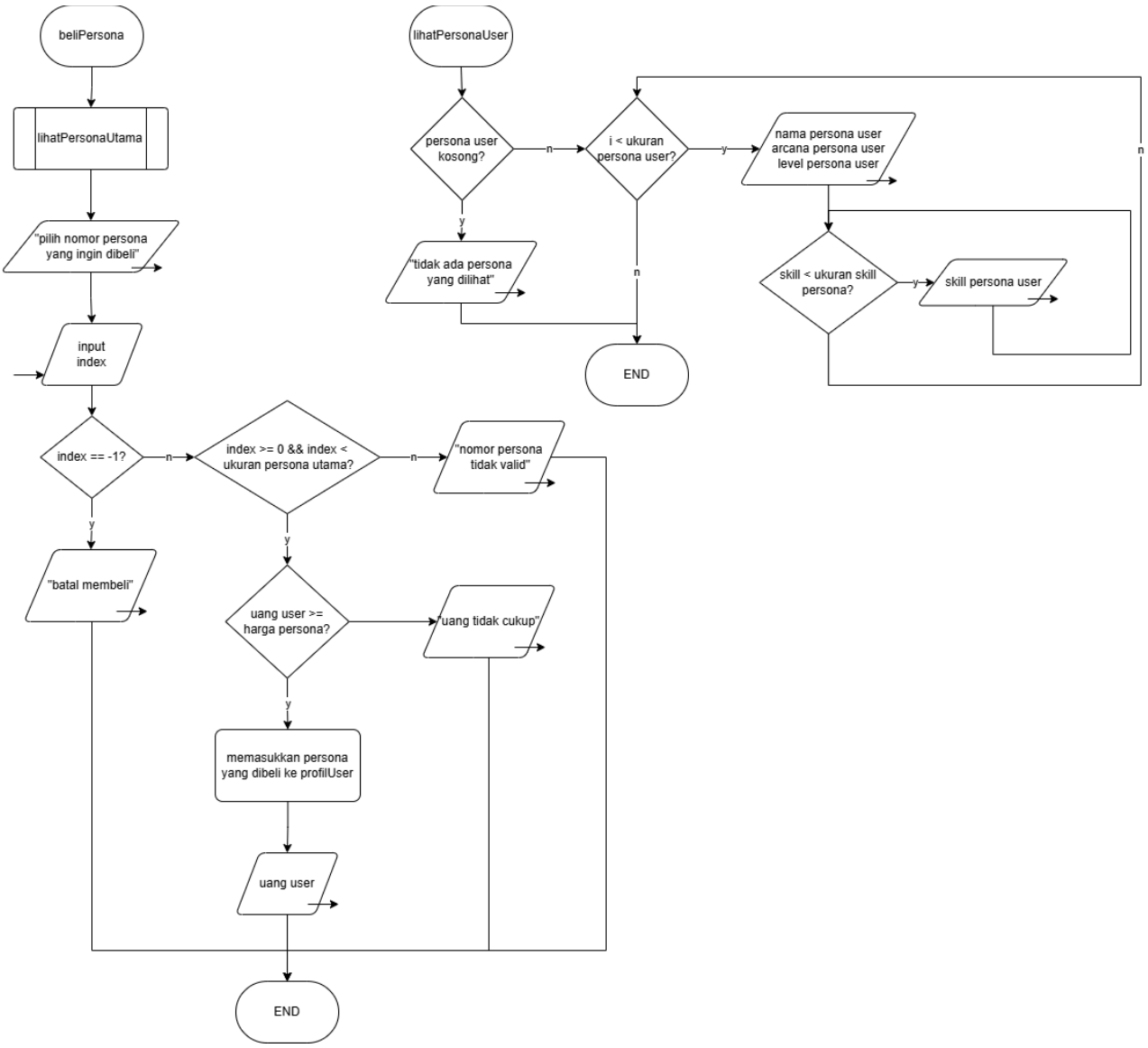
gambar 1.2 flowchart fungsi login dan fungsi register



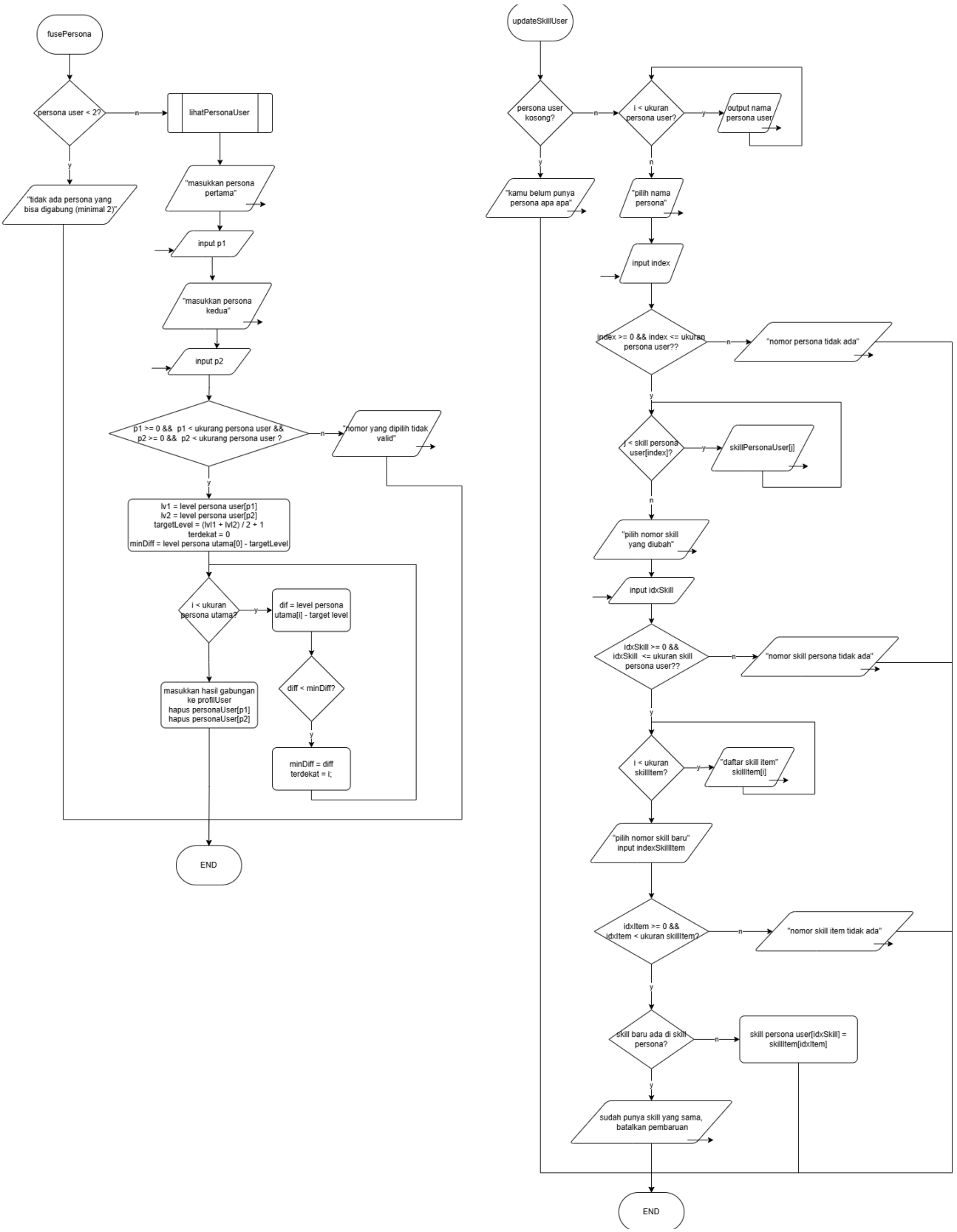
gambar 1.3 flowchart fungsi menuUser



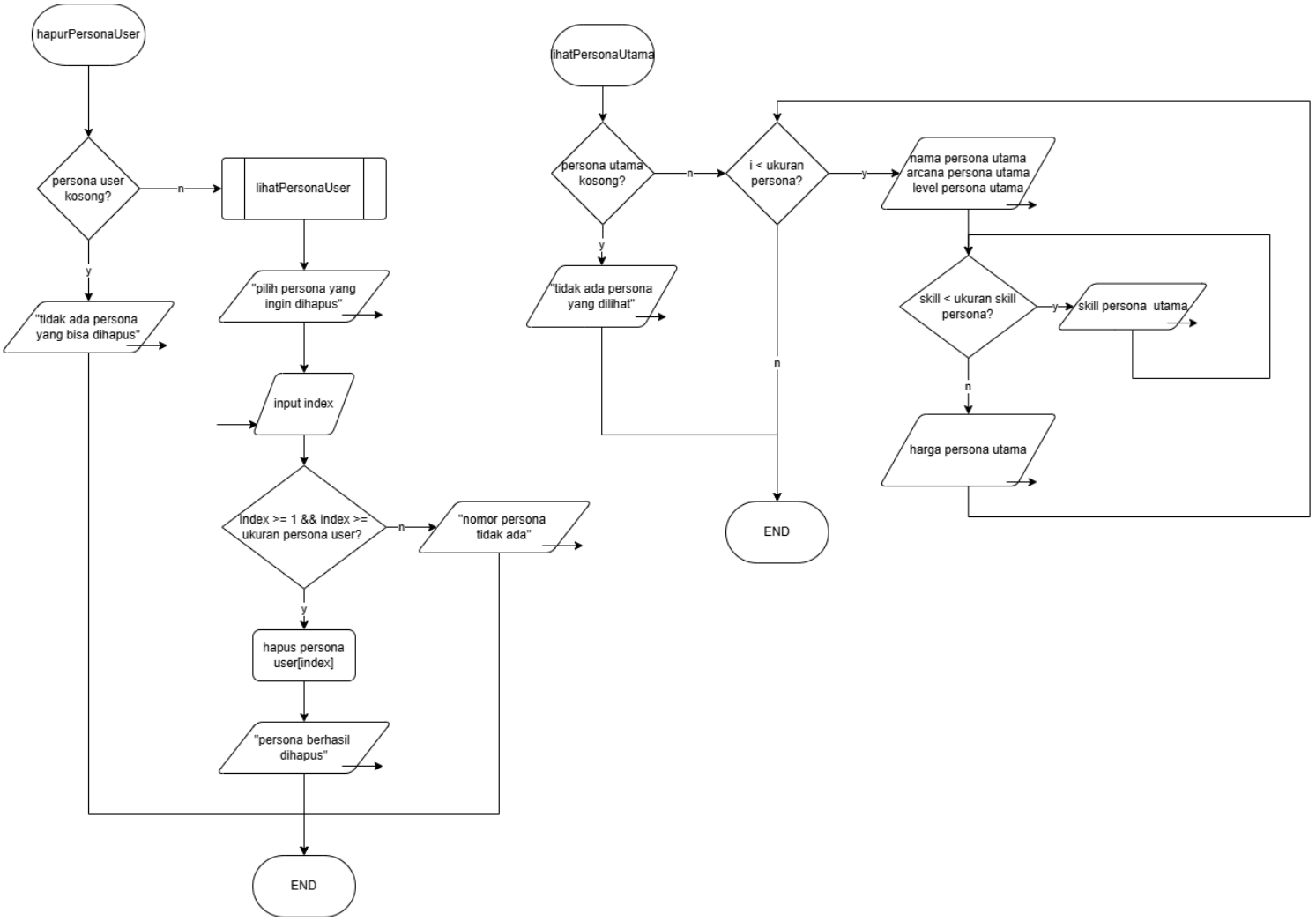
gambar 1.4 flowchart fungsi menuAdmin



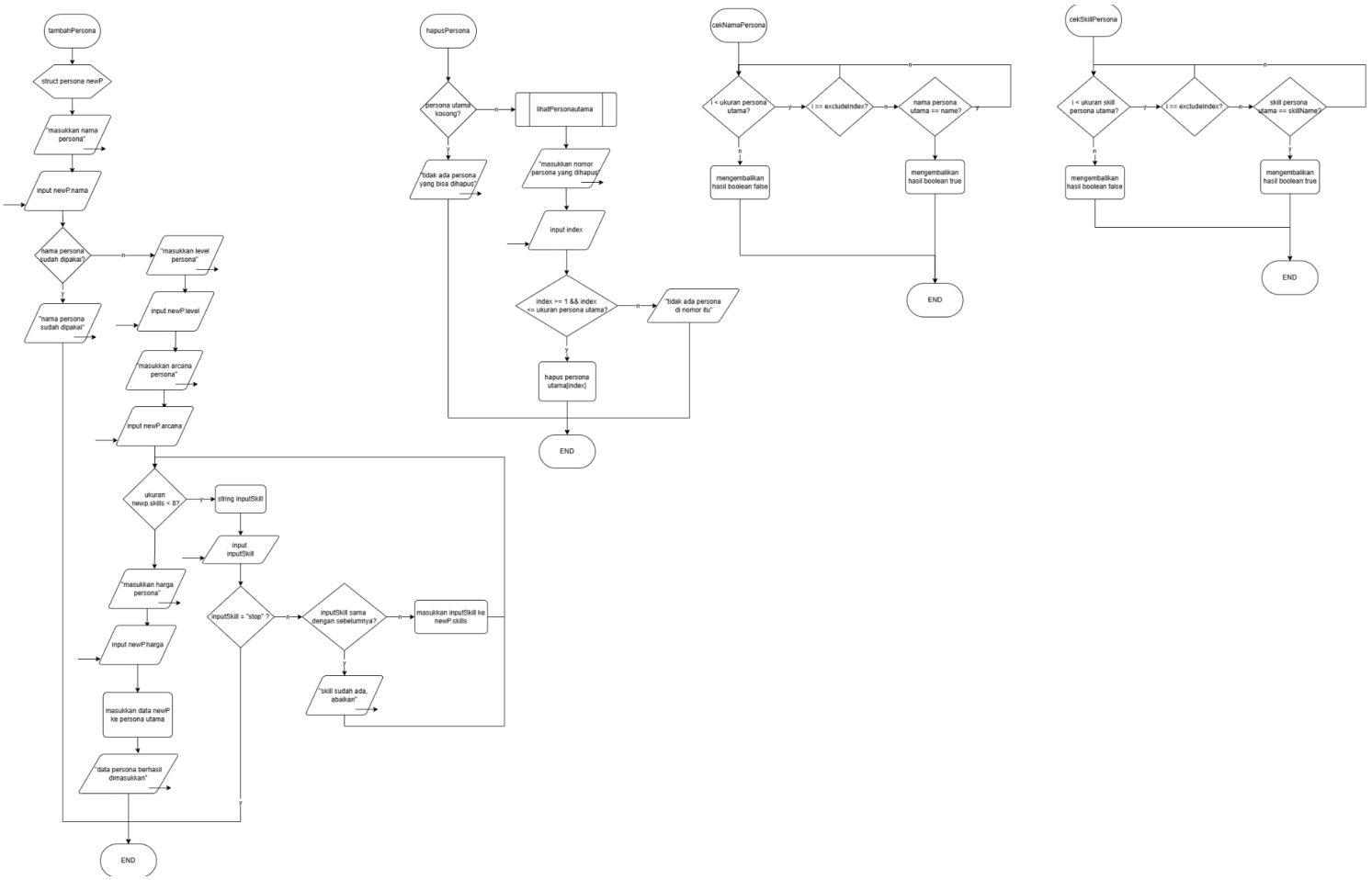
gambar 1.5 flowchart opsi user (fungsi beli dan lihat persona)



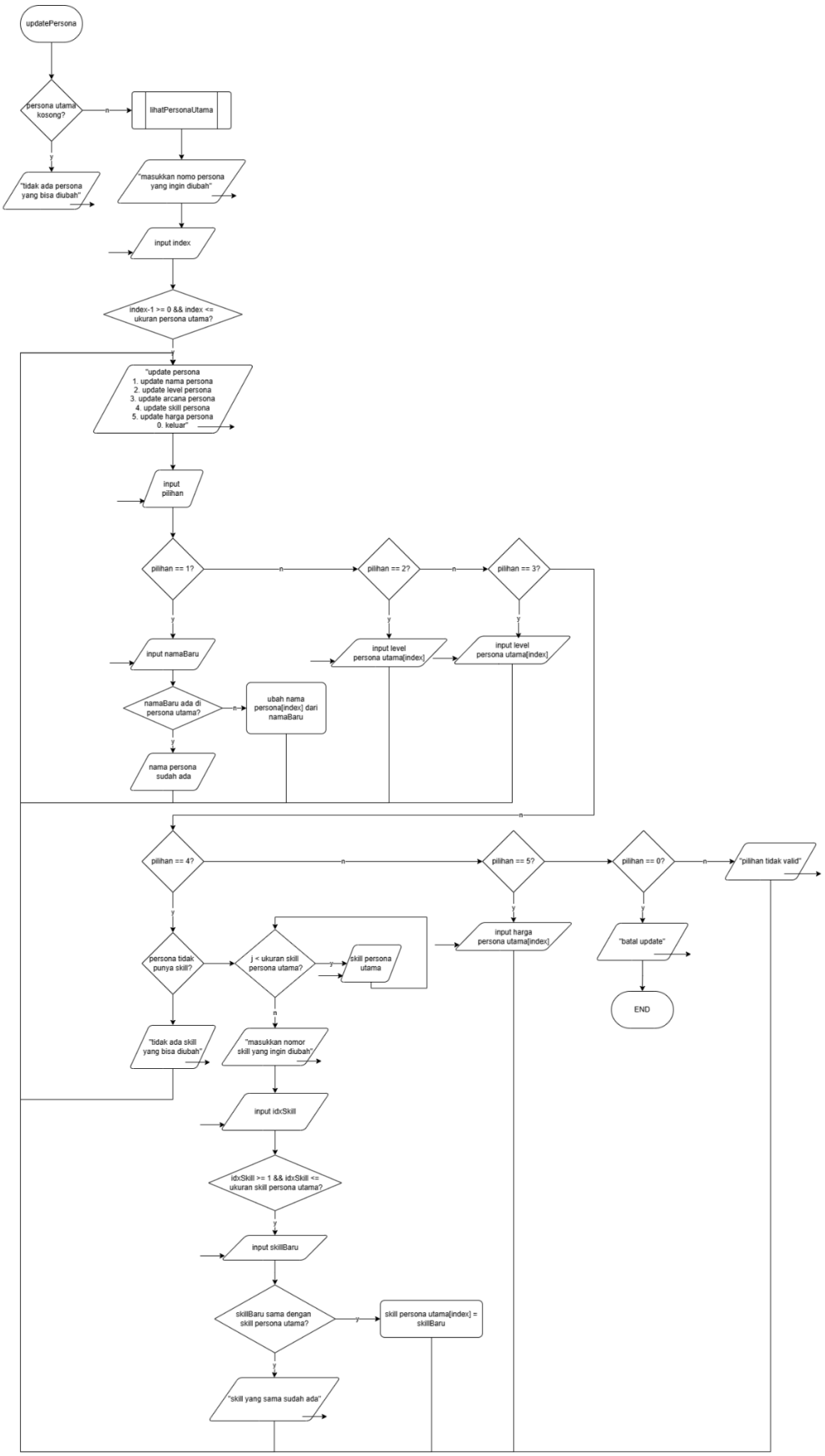
gambar 1.6 opsi user (fungsi gabung dan update persona)



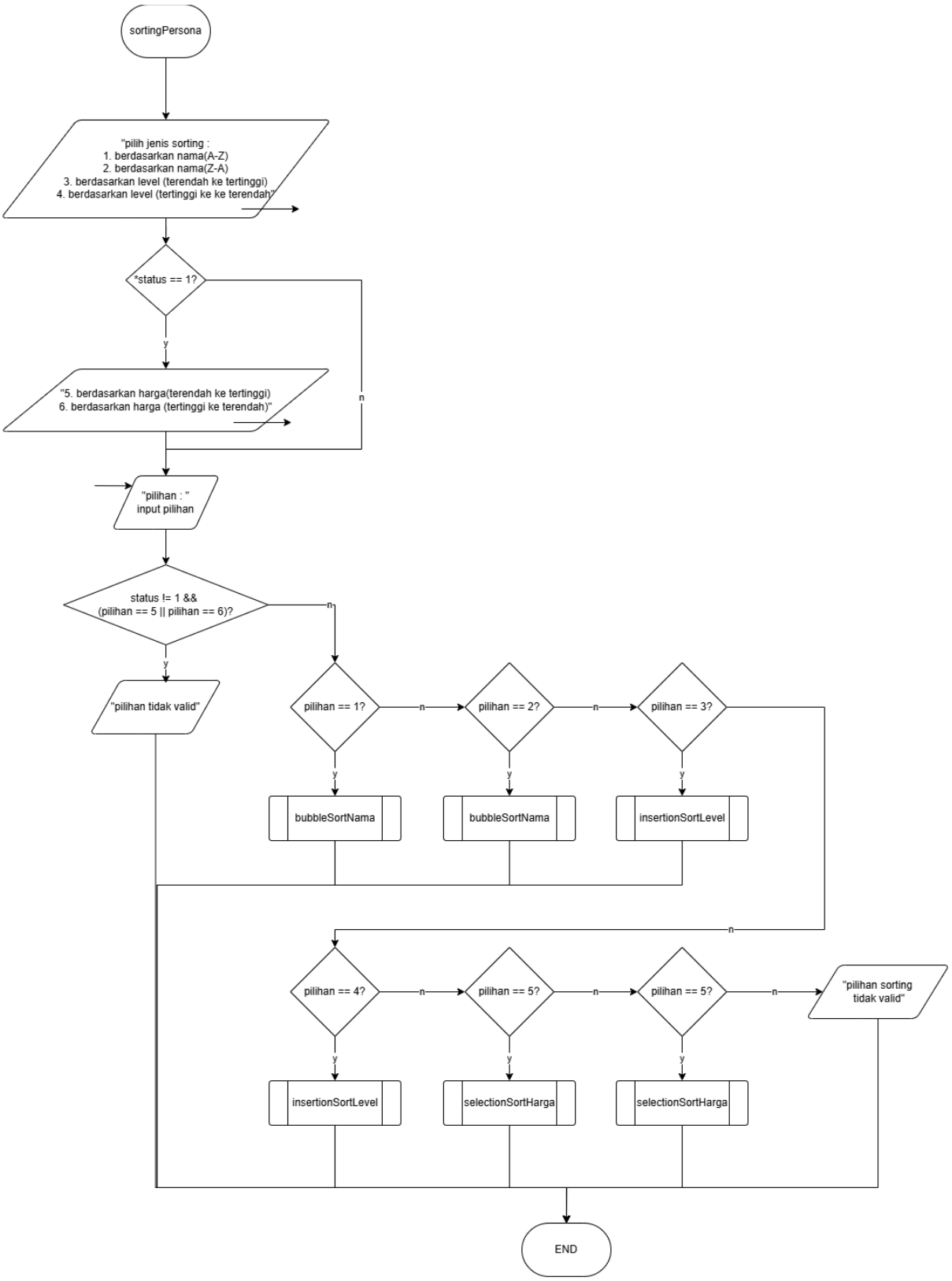
gambar 1.7 opsi user dan admin (fungsi hapus persona user dan fungsi lihat persona utama)



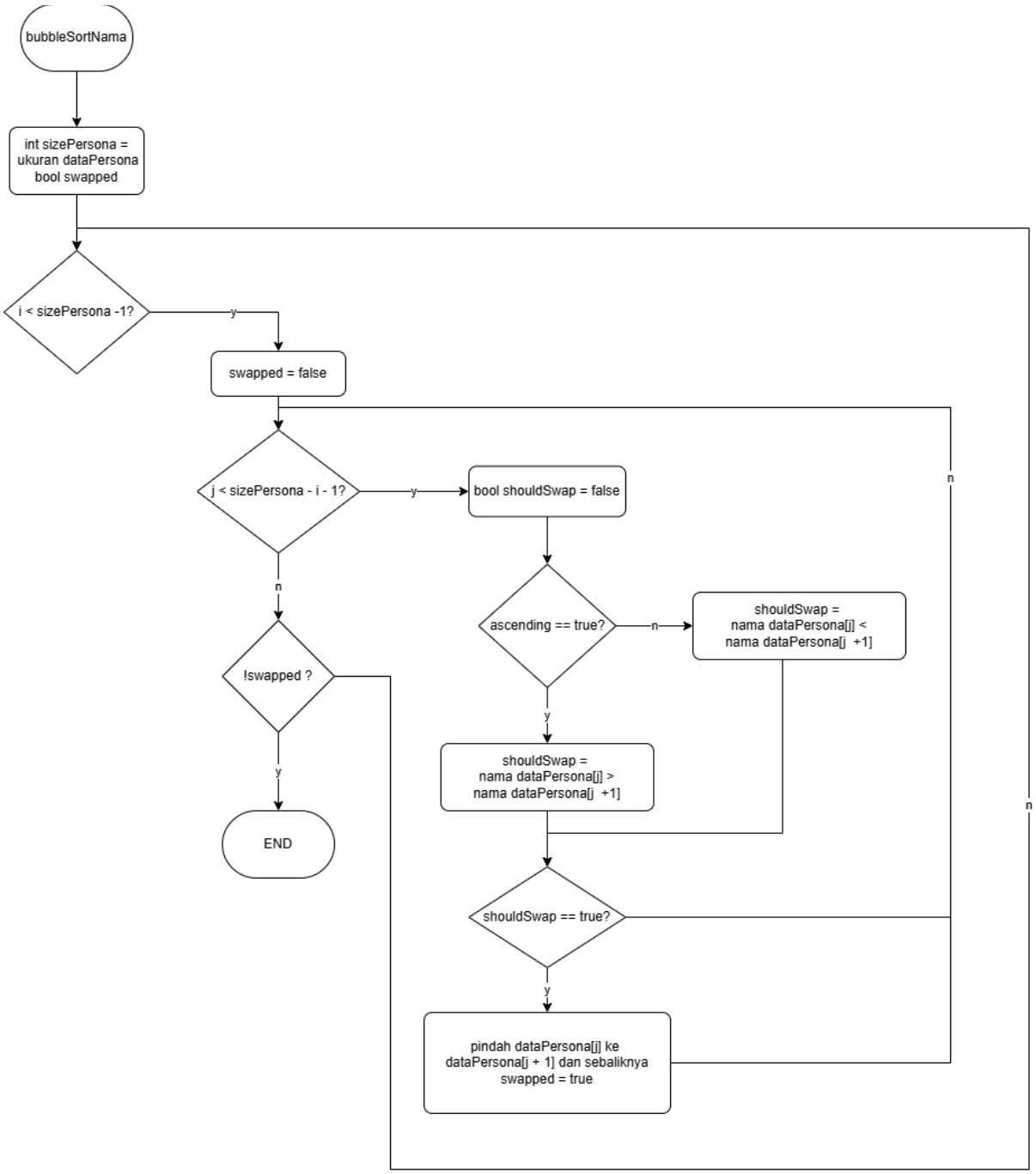
gambar 1.8 opsi admin (fungsi tambah dan hapus persona utama)



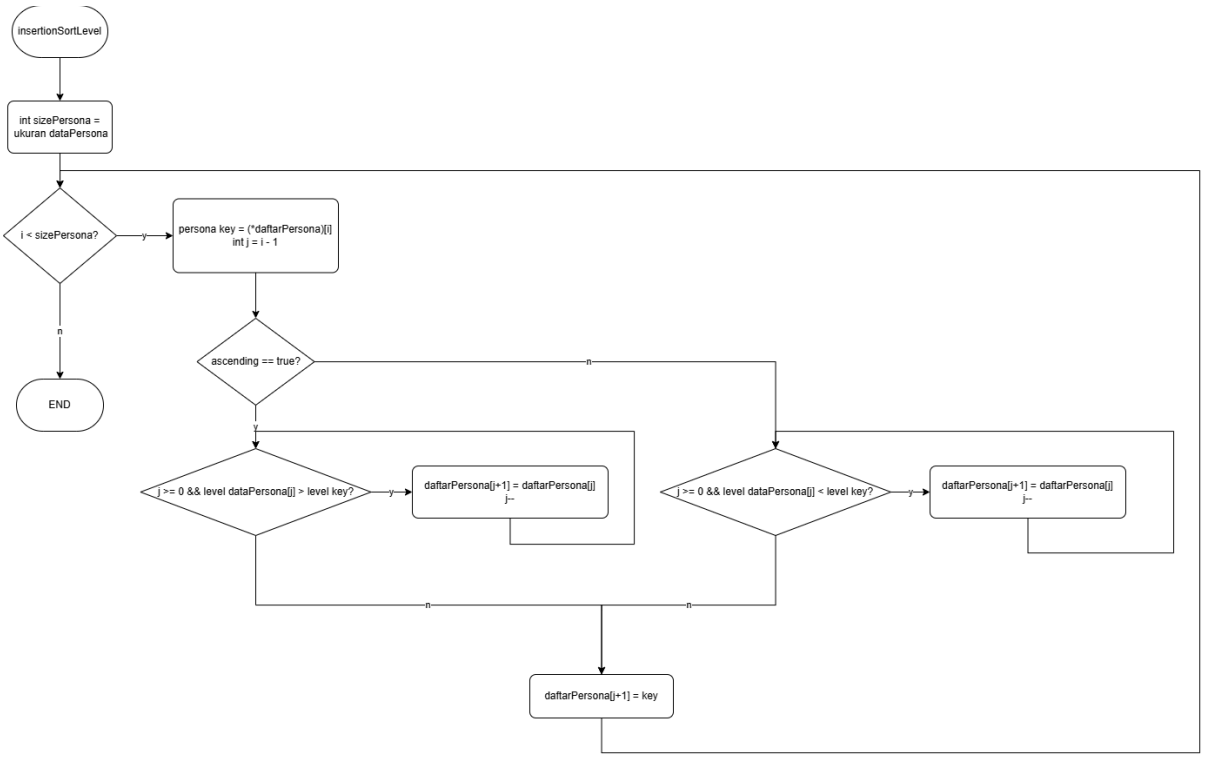
gambar 1.9 opsi admin (update persona utama)



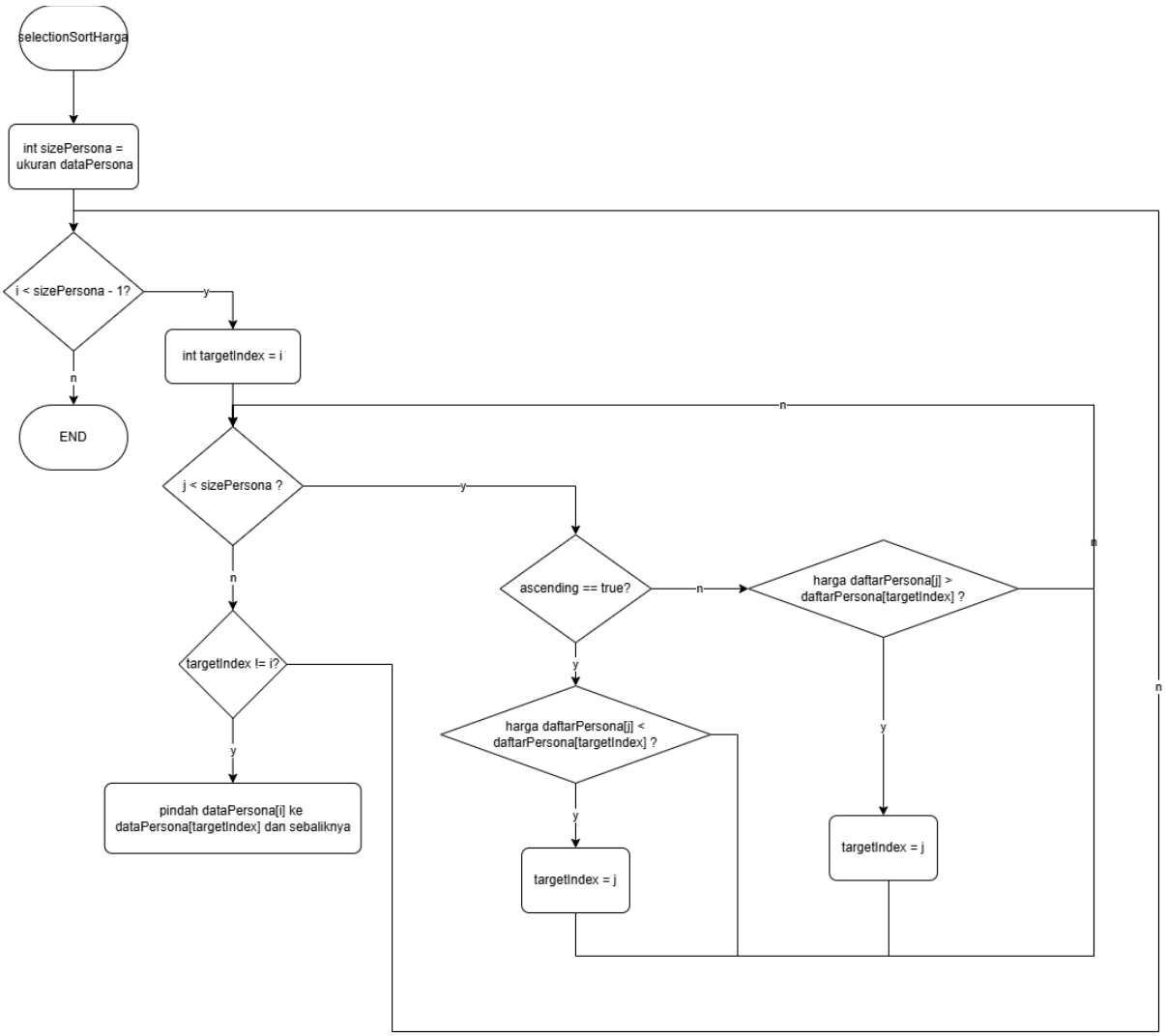
gambar 1.10 opsi user dan admin (menu sorting data persona)



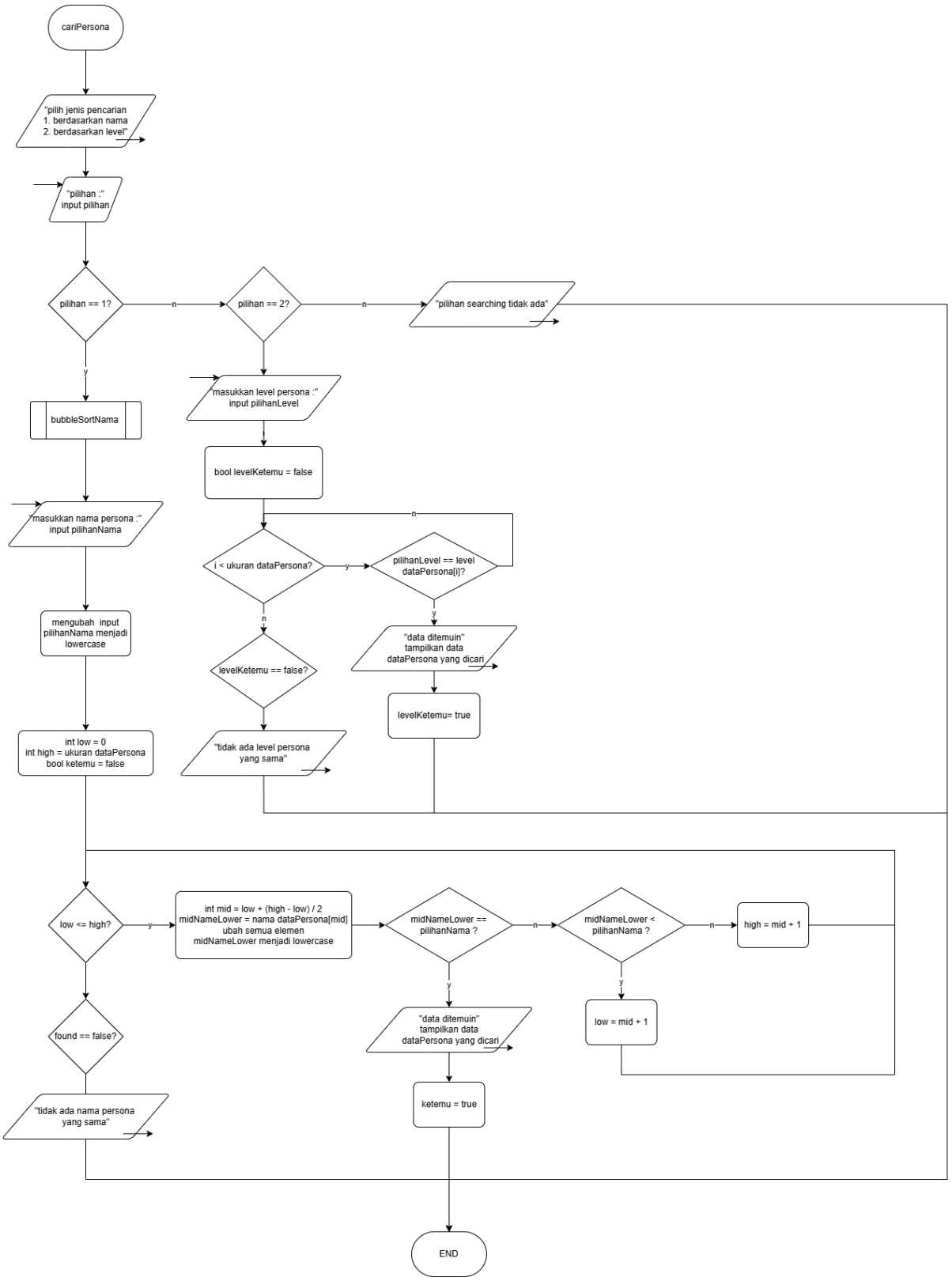
gambar 1.11 bubble sort ascending dan descending



gambar 1.12 insertion sort ascending dan descending



gambar 1.13 selection sort ascending dan descending



1.14 fungsi user dan admin (cari persona)

2. Deskripsi Singkat Program

pada posttest ini saya mecoba membuat searching untuk data persona user dan admin. terdapat 2 cara searching, yaitu berdasarkan nama dan berdasarkan level. binary digunakan pada searching nama dan linear search digunakan untuk searching level.

3. Source Code

A. fitur user

code ini terdapat semua fitur user, mulai dari lihat persona user hingga menghapus nya

Source Code:

```
void lihatPersonaUser(personaUser* profilePtr) {
    if ((*profilePtr).listPersona.empty()) {
        cout << "Tidak ada persona yang dilihat" << endl;
        return;
    }
    for (size_t i = 0; i < (*profilePtr).listPersona.size(); ++i) {
        const auto& p = (*profilePtr).listPersona[i];
        cout << "\n====== "<< i + 1 << p.nama << " ======" << endl;
        cout << "Level    : " << p.level << endl;
        cout << "Arcana   : " << p.arcana << endl;
        cout << "Skill    : ";
        for (const auto& skill : p.skills) {
            cout << skill << ", ";
        }
        cout << endl;
    }
}

void beliPersona(LevelUser* userPtr, personaUser* profilePtr) {
    lihatPersonaUtama();
    int index;
    cout << "Pilih nomor persona yang ingin dibeli (0 untuk batal) : ";
    cin >> index;
    index -= 1;
    if (index == -1 ){
        cout << "batal membeli" << endl;
    }
    if (index >= 0 && index < (int)personaUtama.size()) {
        if ((*userPtr).uang >= personaUtama[index].harga) {
            persona terbeli = personaUtama[index];
            (*profilePtr).listPersona.push_back(terbeli);
```

```
            (*userPtr).uang -= personaUtama[index].harga;
            cout << "persona berhasil dibeli! Sisa uang: " << (*userPtr).uang <<
endl;
        }else{
            cout << "Uang tidak cukup!" << endl;
        }
    } else {
        cout << "Nomor persona tidak valid" << endl;
    }
}

void fusePersona(personaUser* profilePtr) {
    if ((*profilePtr).listPersona.size() < 2) {
        cout << "Tidak ada persona yang bisa digabung (Minimal 2)" << endl;
        return;
    }
    lihatPersonaUser(profilePtr);

    int p1, p2;
    cout << "masukkan nomor persona pertama : "; cin >> p1;
    cout << "masukkan nomor persona kedua : "; cin >> p2;
    p1--; p2--;
    if (p1 >= 0 && p1 < (int)(*profilePtr).listPersona.size() && p2 >= 0 && p2 <
(int)(*profilePtr).listPersona.size() && p1 != p2) {
        int lvl1 = (*profilePtr).listPersona[p1].level;
        int lvl2 = (*profilePtr).listPersona[p2].level;
        int targetLevel = (lvl1 + lvl2) / 2 + 1;
        int terdekat = 0;
        int minDiff = abs(personaUtama[0].level - targetLevel);
        for (size_t i = 1; i < personaUtama.size(); ++i) {
            int diff = abs(personaUtama[i].level - targetLevel);
            if (diff < minDiff) {
                minDiff = diff;
                terdekat = i;
            }
        }
        (*profilePtr).listPersona.push_back(personaUtama[terdekat]);
        cout << "Fusion berhasil! Mendapatkan: " << personaUtama[terdekat].nama <<
endl;
        int maxIdx = max(p1, p2);
        int minIdx = min(p1, p2);
        (*profilePtr).listPersona.erase((*profilePtr).listPersona.begin() + maxIdx);
        (*profilePtr).listPersona.erase((*profilePtr).listPersona.begin() + minIdx);
    } else {
        cout << "Nomor yang dipilih tidak valid" << endl;
    }
}

void updateSkillUser(personaUser* profilePtr) {
    if ((*profilePtr).listPersona.empty()) {
        cout << "Kamu belum punya persona apa-apa" << endl;
        return;
    }
```

```
    for (size_t i = 0; i < (*profilePtr).listPersona.size(); ++i) {
        cout << (i + 1) << ". " << (*profilePtr).listPersona[i].nama << endl;
    }
    int index;
    cout << "Pilih nomor persona : "; cin >> index;
    index -= 1;
    if (index >= 0 && index < (int)(*profilePtr).listPersona.size()) {
        auto& p = (*profilePtr).listPersona[index];
        if (p.skills.empty()) {
            cout << "persona ini tidak memiliki skill" << endl;
            return;
        }
        for (size_t j = 0; j < p.skills.size(); ++j) {
            cout << (j + 1) << ". " << p.skills[j] << endl;
        }
        int idxSkill;
        cout << "Pilih skill yang ingin diubah : "; cin >> idxSkill;
        idxSkill -= 1;
        if (idxSkill >= 0 && idxSkill < (int)p.skills.size()) {
            cout << "Daftar Skill Item:" << endl;
            for (size_t i = 0; i < skillItems.size(); ++i) {
                cout << (i + 1) << ". " << skillItems[i] << endl;
            }
            int idxItem;
            cout << "Pilih nomor skill baru : "; cin >> idxItem;
            idxItem -= 1;
            if (idxItem >= 0 && idxItem < (int)skillItems.size()) {
                string skillBaru = skillItems[idxItem];
                if (cekSkillPersona(p.skills, skillBaru, idxSkill)) {
                    cout << "Sudah punya skill yang sama, batalkan pembaruan" <<
endl;
                } else {
                    p.skills[idxSkill] = skillBaru;
                    cout << "Pembaruan berhasil" << endl;
                }
            } else {
                cout << "Nomor skill item tidak ada" << endl;
            }
        } else {
            cout << "Nomor skill persona tidak ada" << endl;
        }
    } else {
        cout << "Nomor persona tidak ada" << endl;
    }
}

void hapusPersonaUser(personaUser* profilePtr) {
    if ((*profilePtr).listPersona.empty()) {
        cout << "Kamu belum punya persona apa-apa" << endl;
        return;
    }
    lihatPersonaUser(profilePtr);
    int index;
```

```
    cout << "Pilih persona yang ingin dihapus : "; cin >> index;
    if (index >= 1 && index <= (int)(*profilePtr).listPersona.size()) {
        (*profilePtr).listPersona.erase((*profilePtr).listPersona.begin() + (index -
1));
        cout << "persona berhasil dihapus" << endl;
    } else {
        cout << "Nomor persona tidak ada" << endl;
    }
}

void userMenu(int userIndex) {
    int profilIndex = cariAtauBuatProfil(users[userIndex].nama);

    LevelUser* currentUserPtr = &users[userIndex];
    personaUser* currentUserProfilePtr = &profilUser[profilIndex];

    int pilihan;
    do {
        cout << "\n= Menu Velvet Room User =" << endl;
        cout << "User: " << currentUserPtr->nama << " | Uang: " <<
currentUserPtr->uang << " yen" << endl;
        cout << "1. Lihat data persona user" << endl;
        cout << "2. cari persona" << endl;
        cout << "3. Beli persona" << endl;
        cout << "4. Fuse/Gabung persona" << endl;
        cout << "5. Ubah skill persona" << endl;
        cout << "6. sorting persona" << endl;
        cout << "7. Hapus persona" << endl;
        cout << "0. Keluar" << endl;
        cout << "pilihan : ";
        cin >> pilihan;
        switch (pilihan) {

            case 1: lihatPersonaUser(currentUserProfilePtr); break;
            case 2: cariPersona(&(currentUserProfilePtr->listPersona),
&(currentUserPtr->status)); break;
            case 3: beliPersona(currentUserPtr, currentUserProfilePtr); break;
            case 4: fusePersona(currentUserProfilePtr); break;
            case 5: updateSkillUser(currentUserProfilePtr); break;
            case 6: sortingPersona(&(currentUserProfilePtr->listPersona),
&(currentUserPtr->status)); break;
            case 7: hapusPersonaUser(currentUserProfilePtr); break;
            case 0: cout << "Log out" << endl; break;
            default: cout << "pilihan tidak valid." << endl;
        }
    } while (pilihan != 0);
}
```

B. Fitur admin

pada kode ini terdapat semua fitur admin, mulai dari lihat data utama persona, hingga hapus persona dari data utama

```
void lihatPersonaUtama() {
    if (personaUtama.empty()) {
        cout << "Tidak ada persona yang bisa dibaca" << endl;
        return;
    }
    for (size_t i = 0; i < personaUtama.size(); ++i) {
        cout << "\n====== " << i + 1 << personaUtama[i].nama << " ======" << endl;
        cout << "Level    : " << personaUtama[i].level << endl;
        cout << "Arcana   : " << personaUtama[i].arcana << endl;
        cout << "Skill    : ";
        for (const auto& skill : personaUtama[i].skills) {
            cout << skill << ", ";
        }
        cout << endl;
        cout << "Harga    : " << personaUtama[i].harga << endl;
    }
}

void tambahPersona() {

    persona newP;
    cout << "masukkan nama persona : ";
    clearInputBuffer();
    getline(cin, newP.nama);

    if (cekNamaPersona(newP.nama)) {
        cout << "Gagal: Nama persona sudah ada!" << endl;
        return;
    }

    cout << "masukkan level persona : ";
    cin >> newP.level;
    cout << "masukkan arcana persona : ";
    cin >> newP.arcana;

    cout << "masukkan skill persona (ketik 'stop' untuk berhenti):" << endl;
    clearInputBuffer();
    while (newP.skills.size() < 8) {
        string inputSkill;
        cout << "Skill " << (newP.skills.size() + 1) << ": ";
        getline(cin, inputSkill);
        if (inputSkill == "stop") break;

        if (cekSkillPersona(newP.skills, inputSkill)) {
            cout << "Skill tersebut sudah ada di persona ini, abaikan." << endl;
        } else {
            newP.skills.push_back(inputSkill);
        }
    }
```

```
    cout << "masukkan harga persona : ";
    cin >> newP.harga;

    personaUtama.push_back(newP);
    cout << "Data persona berhasil dimasukkan" << endl;
}

void updatePersona() {
    if (personaUtama.empty()) {
        cout << "Tidak ada persona yang bisa diubah" << endl;
        return;
    }

    lihatPersonaUtama();

    int index;
    cout << "masukkan nomor persona yang ingin diubah: ";
    cin >> index;

    if (index - 1 >= 0 && index - 1 < (int)personaUtama.size()) {
        int idx = index - 1;
        int pilihan;
        do {
            cout << "\n--- Update persona: " << personaUtama[idx].nama << " ---" <<
endl;
            cout << "1. Ubah nama persona" << endl;
            cout << "2. Ubah level persona" << endl;
            cout << "3. Ubah arcana persona" << endl;
            cout << "4. Ubah skill persona" << endl;
            cout << "5. Ubah harga persona" << endl;
            cout << "0. Batal update" << endl;
            cout << "pilihan: ";
            cin >> pilihan;

            switch (pilihan) {
            case 1: {
                string namaBaru;
                cout << "masukkan nama baru : ";
                clearInputBuffer();
                getline(cin, namaBaru);
                if (cekNamaPersona(namaBaru, idx)) {
                    cout << "Gagal: Nama persona sudah ada!" << endl;
                } else {
                    personaUtama[idx].nama = namaBaru;
                    cout << "Berhasil diubah" << endl;
                }
                break;
            }
            case 2:
                cout << "masukkan level baru : ";
                cin >> personaUtama[idx].level;
                cout << "Berhasil diubah" << endl;
```

```
                break;
            case 3:
                cout << "masukkan arcana baru : ";
                cin >> personaUtama[idx].arcana;
                cout << "Berhasil diubah" << endl;
                break;
            case 4: {
                if (personaUtama[idx].skills.empty()) {
                    cout << "Tidak ada skill untuk diubah" << endl;
                    break;
                }
                for (size_t j = 0; j < personaUtama[idx].skills.size(); ++j) {
                    cout << "Skill ke " << (j + 1) << " : " <<
personaUtama[idx].skills[j] << endl;
                }
                int idxSkill;
                cout << "masukkan nomor skill yang ingin diubah : ";
                cin >> idxSkill;
                if (idxSkill >= 1 && idxSkill <=
(int)personaUtama[idx].skills.size()) {
                    clearInputBuffer();
                    string skillBaru;
                    cout << "masukkan skill baru : ";
                    getline(cin, skillBaru);

                    if (cekSkillPersona(personaUtama[idx].skills, skillBaru,
idxSkill - 1)) {
                        cout << "Gagal: Skill sudah ada di persona ini!" << endl;
                    } else {
                        personaUtama[idx].skills[idxSkill - 1] = skillBaru;
                        cout << "Skill berhasil diubah" << endl;
                    }
                } else {
                    cout << "Nomor skill tidak valid" << endl;
                }
                break;
            }
            case 5:
                cout << "masukkan harga baru : ";
                cin >> personaUtama[idx].harga;
                cout << "Berhasil diubah" << endl;
                break;
            case 0:
                cout << "Batal update" << endl;
                break;
            default:
                cout << "pilihan tidak valid" << endl;
            }
        } while (pilihan != 0);

    }else{
        cout << "Nomor tidak valid" << endl;
        return;
```

```
    }

}

void hapusPersona() {
    if (personaUtama.empty()) {
        cout << "Tidak ada persona yang bisa dihapus" << endl;
        return;
    }

    lihatPersonaUtama();

    int index;
    cout << "masukkan nomor persona yang ingin dihapus : ";
    cin >> index;

    if (index >= 1 && index <= (int)personaUtama.size()) {
        personaUtama.erase(personaUtama.begin() + (index - 1));
        cout << "persona berhasil dihapus" << endl;
    } else {
        cout << "Tidak ada persona di nomor itu" << endl;
    }
}

void menuAdmin(LevelUser* userPtr) {
    int pilihan;
    do {
        cout << "\n= Menu Velvet Room Admin =" << endl;
        cout << "1. Tampilkan data persona" << endl;
        cout << "2. Cari persona" << endl;
        cout << "3. Tambah persona" << endl;
        cout << "4. Update persona" << endl;
        cout << "5. Hapus persona" << endl;
        cout << "6. sorting persona" << endl;
        cout << "0. Keluar" << endl;
        cout << "pilihan: ";
        cin >> pilihan;
        switch (pilihan) {
            case 1: lihatPersonaUtama(); break;
            case 2: cariPersona(&personaUtama, &(userPtr->status)); break;
            case 3: tambahPersona(); break;
            case 4: updatePersona(); break;
            case 5: hapusPersona(); break;
            case 6: sortingPersona(&personaUtama, &(userPtr->status)); break;
            case 0:
                cout << "Log out.... " << (*userPtr).nama << endl;
                break;
            default: cout << "pilihan tidak valid." << endl;
        }
    } while (pilihan != 0);
}
```

C. Fitur searching persona

pada kode ini terdapat fitur baru, yaitu searching, terdapat 2 searching yang bisa dipakai, yaitu searching berdasarkan nama atau berdasarkan level

```
void cariPersona(vector<persona>* daftarPersona, int* status) {
    int pilihan;
    cout << "\npilih jenis pencarian : " << endl;
    cout << "1. berdasarkan nama" << endl;
    cout << "2. berdasarkan level" << endl;
    cout << "pilihan : ";
    cin >> pilihan;

    switch (pilihan) {
        case 1: {
            string pilihanNama;
            bubbleSortNama(daftarPersona, true);
            cout << "masukkan nama persona : ";
            clearInputBuffer();
            getline(cin, pilihanNama);

            std::transform(pilihanNama.begin(), pilihanNama.end(),
pilihanNama.begin(),
                           [](unsigned char c){ return std::tolower(c); });

            int low = 0;
            int high = daftarPersona->size() - 1;
            bool ketemu = false;
            while (low <= high) {
                int mid = low + (high - low) / 2;

                string midNameLower = (*daftarPersona)[mid].nama;
                std::transform(midNameLower.begin(), midNameLower.end(),
midNameLower.begin(),
                               [](unsigned char c){ return std::tolower(c);
});

                if (midNameLower == pilihanNama) {
                    cout << "data ditemuin " << endl;
                    cout << "\n====== " << (*daftarPersona)[mid].nama << "
======" << endl;
                    cout << "Level    : " << (*daftarPersona)[mid].level <<
endl;
                    cout << "Arcana   : " << (*daftarPersona)[mid].arcana <<
endl;
                    cout << "Skill    : ";
                    for (const auto& skill : (*daftarPersona)[mid].skills) {
                        cout << skill << ", ";
```

```
                    }
                    cout << endl;
                    if (*status == 1){
                        cout << "harga    : " << (*daftarPersona)[mid].harga
<< endl;
                    }
                    ketemu = true;
                    break;
                } else if (midNameLower < pilihanNama) {
                    low = mid + 1;
                } else {
                    high = mid - 1;
                }
            }
            if (!ketemu) {
                cout << "tidak ada nama persona yang sama " << endl;
            }
            break;
        }

        case 2: {
            int pilihanLevel;
            cout << "masukkan level persona yang dicari : ";
            cin >> pilihanLevel;

            bool levelKetemu = false;
            for (int i = 0; i < daftarPersona->size(); i++) {
                if (pilihanLevel == (*daftarPersona)[i].level) {
                    cout << "data ditemuin " << endl;
                    cout << "\n====== " << (*daftarPersona)[i].nama << "
======" << endl;
                    cout << "Level    : " << (*daftarPersona)[i].level <<
endl;
                    cout << "Arcana   : " << (*daftarPersona)[i].arcana <<
endl;
                    cout << "Skill    : ";
                    for (const auto& skill : (*daftarPersona)[i].skills) {
                        cout << skill << ", ";
                    }
                    cout << endl;
                    levelKetemu = true;
                }
            }
            if (!levelKetemu) {
                cout << "tidak ada level persona yang sama" << endl;
            }
            break;
        }
```

```
        default: {
            cout << "pilihan searching tidak valid" << endl;
            break;
        }
    }
}
```

4. Hasil Output

```
pilihan : 2

====== 1orpheus ======
Level     : 10
Arcana    : fool
Skill     : agi, dia, bash, tarunda,
Harga     : 500

====== 2jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga     : 750

====== 3archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Harga     : 1000
Pilih nomor persona yang ingin dibeli (0 untuk batal) : 1
persona berhasil dibeli! Sisa uang: 9500
```

```
====== 1orpheus ======
Level     : 10
Arcana    : fool
Skill     : agi, dia, bash, tarunda,
```

Gambar 4.1 opsi user(beli persona)

```
pilihan : 3

====== 1orpheus ======
Level     : 10
Arcana    : fool
Skill     : agi, dia, bash, tarunda,

====== 2archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
masukkan nomor persona pertama : 1
masukkan nomor persona kedua : 2
Fusion berhasil! Mendapatkan: jack o lantern
```

```
====== 1jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
```

Gambar 4.2 opsi user (fuse/gabung persona)

```
pilihan : 4
1. jack o lantern
Pilih nomor persona : 1
1. bufu
2. pulinpa
3. mabufu
4. shock punch
5. freeze boost
6. magic boost
Pilih skill yang ingin diubah : 2
Daftar Skill Item:
1. masukunda
2. dekaja
3. debilitate
4. charge
5. concentrate
Pilih nomor skill baru : 3
Pembaruan berhasil
```

```
====== 1jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, debilitate, mabufu, shock punch, freeze boost, magic boost,
```

gambar 4.3 opsi user (ubah skill persona)

```
pilihan : 5

====== 1jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, debilitate, mabufu, shock punch, freeze boost, magic boost,

====== 2archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Pilih persona yang ingin dihapus : 1
persona berhasil dihapus
```

```
pilihan : 1

====== 1archangel ======
Level     : 20
Arcana    : justice
Skill     : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
```

gambar 4.4 opsi user (hapus persona user)

```
pilihan: 2
masukkan nama persona : siegfried
masukkan level persona : 62
masukkan arcana persona : strength
masukkan skill persona (ketik 'stop' untuk berhenti):
Skill 1: charge
Skill 2: brave blade
Skill 3: crit rate amp
Skill 4: amp slash
Skill 5: stop
masukkan harga persona : 5000
Data persona berhasil dimasukkan
```

```
====== 4siegfried ======
Level     : 62
Arcana    : strength
Skill     : charge, brave blade, crit rate amp, amp slash,
Harga     : 5000
```

gambar 4.5 opsi admin (tambah persona)

```
--- Update persona: jack o lantern ---
1. Ubah nama persona
2. Ubah level persona
3. Ubah arcana persona
4. Ubah skill persona
5. Ubah harga persona
0. Batal update
pilihan: 4
Skill ke 1 : bufu
Skill ke 2 : pulinpa
Skill ke 3 : mabufu
Skill ke 4 : shock punch
Skill ke 5 : freeze boost
Skill ke 6 : magic boost
masukkan nomor skill yang ingin diubah : 1
masukkan skill baru : mabufudyne
Skill berhasil diubah
```

```
====== 2jack o lantern ======
Level    : 15
Arcana   : magician
Skill    : mabufudyne, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga    : 750
```

gambar 4.6 opsi admin (edit persona)

```
pilihan: 4

====== 1orpheus ======
Level    : 10
Arcana   : fool
Skill    : agi, dia, bash, tarunda,
Harga    : 500

====== 2jack o lantern ======
Level    : 15
Arcana   : magician
Skill    : mabufudyne, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga    : 750

====== 3archangel ======
Level    : 20
Arcana   : justice
Skill    : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Harga    : 1000

====== 4siegfried ======
Level    : 62
Arcana   : strength
Skill    : charge, brave blade, crit rate amp, amp slash,
Harga    : 5000
masukkan nomor persona yang ingin dihapus : 2
persona berhasil dihapus
```

```
====== 1orpheus ======
Level    : 10
Arcana   : fool
Skill    : agi, dia, bash, tarunda,
Harga    : 500

====== 2archangel ======
Level    : 20
Arcana   : justice
Skill    : kouha, single shot, baisudi, sukukaja, holy arrow, apt pupil,
Harga    : 1000

====== 3siegfried ======
Level    : 62
Arcana   : strength
Skill    : charge, brave blade, crit rate amp, amp slash,
Harga    : 5000
```

gambar 4.7 opsi admin(hapus persona)

```
 pilih jenis sorting :
1. berdasarkan nama(A-Z)
2. berdasarkan nama(Z-A)
3. berdasarkan level(terendah ke tertinggi)
4. berdasarkan level(tertinggi ke terendah)
5. berdasarkan harga(termurah ke termahal)
6. berdasarkan harga(termahal ke termurah)
pilihan :
1
```

```
====== 1jack o lantern ======
Level    : 15
Arcana   : magician
Skill    : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga    : 750

====== 2orpheus ======
Level    : 10
Arcana   : fool
Skill    : agi, dia, bash, tarunda,
Harga    : 500

====== 3sigfried ======
Level    : 62
Arcana   : strength
Skill    : charge, brave blade, crite rate amp, amp slash,
Harga    : 5000
```

gambar 4.8 opsi sorting persona (berdasarkan nama)

```
 pilih jenis sorting :
1. berdasarkan nama(A-Z)
2. berdasarkan nama(Z-A)
3. berdasarkan level(terendah ke tertinggi)
4. berdasarkan level(tertinggi ke terendah)
5. berdasarkan harga(termurah ke termahal)
6. berdasarkan harga(termahal ke termurah)
pilihan :
5
```

```
====== 1orpheus ======
Level     : 10
Arcana    : fool
Skill     : agi, dia, bash, tarunda,
Harga     : 500

====== 2jack o lantern ======
Level     : 15
Arcana    : magician
Skill     : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
Harga     : 750

====== 3sigfried ======
Level     : 62
Arcana    : strength
Skill     : charge, brave blade, crite rate amp, amp slash,
Harga     : 5000
```

gambar 4.9 sorting persona (berdasarkan harga)

```
pilih jenis pencarian :
1. berdasarkan nama
2. berdasarkan level
pilihan : 1
masukkan nama persona : thanatos
data ditemuin

====== thanatos ======
Level     : 78
Arcana    : death
Skill     : maeigaon, mamudoon, ghastly wail,
harga     : 45000
```

gambar 4.10 cari persona (berdasarkan nama)

```
pilih jenis pencarian :
1. berdasarkan nama
2. berdasarkan level
pilihan : 2
masukkan level persona yang dicari : 15
data ditemuin

====== jack o lantern ======
Level    : 15
Arcana   : magician
Skill    : bufu, pulinpa, mabufu, shock punch, freeze boost, magic boost,
```

gambar 4.11 cari persona (berdasarkan level)

```
=== SELAMAT DATANG DI VELVET ROOM ===
1. login
2. register
0. keluar
pilihan : 2

=== Registrasi user baru ===
masukkan nama user : aigis
masukkan password : robot123
registrasi berhasil
```

```
--- Menu Login ---
masukkan nama (atau 0 untuk keluar): aigis
masukkan Password : robot123
selamat datang aigis

= Menu Velvet Room User =
User: aigis | Uang: 10000 yen
```

gambar 4.12 register user

5. Langkah-langkah GIT

5.1 GIT Init

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git init
Reinitialized existing Git repository in G:/praktikum/APL/.git/
```

gambar 5.1 inisiasi git

5.2 GIT Add

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git add /g/praktikum/APL/kelas/post-test/post-test-apl-6/2509106038-FEBRIANNO-
OZORA-ALINANTO-PT-6.cpp /g/praktikum/APL/kelas/post-test/post-test-apl-6/2509106
038-FEBRIANNO-OZORA-ALINANTO-PT-6.exe
```

gambar 5.2 menambahkan file baru ke git

5.3 GIT Commit

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git commit -m "bunch of mess part 5"
[main e094e04] bunch of mess part 5
 2 files changed, 772 insertions(+)
 create mode 100644 kelas/post-test/post-test-apl-6/2509106038-FEBRIANNO-OZORA-A
LINANTO-PT-6.cpp
 create mode 100644 kelas/post-test/post-test-apl-6/2509106038-FEBRIANNO-OZORA-A
LINANTO-PT-6.exe
```

gambar 5.3 commit git local

5.4 GIT Remote

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git remote add origin https://github.com/ojola444/praktikum-apl/git
error: remote origin already exists.
```

gambar 5.4 git remote

5.5 GIT Push

```
ozora@ozra MINGW64 /g/praktikum/APL (main)
$ git push -u origin main
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (7/7), 168.81 KiB | 8.04 MiB/s, done.
Total 7 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
To https://github.com/ojola444/praktikum-apl.git
   caf4960..e094e04  main -> main
branch 'main' set up to track 'origin/main'.
```

gambar 5.5 push commit local ke repository main

5.6 hasil push github

2509106038-FEBRIANNO-OZORA-ALINANTO-PT-6.cpp	bunch of mess part 5	12 minutes ago
2509106038-FEBRIANNO-OZORA-ALINANTO-PT-6.exe	bunch of mess part 5	12 minutes ago

gambar 5.6 hasil file yang di push di github